Chapter 10: The Other Side of Zero (Integers)

Worksheet 10: Understanding Integers

Learning Objective: Students will understand the need for integers; represent integers on a number line; compare and order integers; and understand the concept of absolute value.

Worked Example

Represent -3 and $+5$ on a number line and compare them.
Solution: On the number line, numbers increase from left to right.
-3 is to the left of 0, and $+5$ is to the right. Therefore, $-3 < +5$.
$|-3| = 3$ (absolute value); $|+5| = 5$.

Questions 1–5: Easy / Recall

Q1. Write the opposite (additive inverse) of: $+7$, -12, 0, -25.

Q2. Which is greater: -8 or -3?

Q3. Arrange in ascending order: $-5, 3, -1, 7, 0, -9$.

Q4. Find: $|-15|$ and $|+8|$.

Q5. Write an integer for: *15°C below zero*, *a profit of 200*, *200 m below sea level.*

Questions 6–10: Basic Application

Q6. $(-5) + (-3) =$

Q7. $(+8) + (-12) =$

Q8. $(-7) - (-4) =$

Q9. $(-6) \times (+5) =$

Q10. $(-18) \div (-3) =$

Questions 11–15: Practice and Skill Building

Q11. $(-4) + (+9) + (-6) + (+2) =$

Q12. $(-8) \times (-7) =$

Q13. $(+24) \div (-6) =$

Q14. If $a = -3$ and $b = 5$, find $a + b$, $a - b$, $a \times b$, $a \div b$ (as a fraction).

Q15. Complete the table:

a	b	$a \times b$	$a \div b$
$+6$	-4		
-9	-3		
-5	$+7$		

Questions 16–18: Higher Order Thinking

Q16. The temperature at noon in Shimla is -2C. By evening it drops 8C more. By midnight it rises 5C. What is the midnight temperature?

Q17. A submarine is at -200 m. It rises 75 m and then dives 120 m. What is its final depth?

Q18. Find the sum of all integers from -10 to $+10$. What pattern do you notice?

Questions 19–20: Real-Life Applications

Q19. A shopkeeper makes a profit of 350 on Monday, a loss of 120 on Tuesday, a loss of 80 on Wednesday, and a profit of 200 on Thursday. Find his net profit or loss over the four days.

Q20. A town recorded temperatures (°C) for a week: Mon -3, Tue -1, Wed 2, Thu 5, Fri -2, Sat 4, Sun 1. Find the average temperature. On how many days was the temperature below average?

Reflection Question: Why do we need negative numbers? Give three real-life situations where negative numbers are useful.